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Multi-modal contrastive learning based on molecular and textual data for drug response prediction

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ABSTRACT

Accurate prediction of drug response is crucial for advancing precision medicine and optimizing therapeutic strategies. However, current deep learning-based drug response prediction (DRP) methods still suffer from significant limitations. They predominantly rely on molecular features while overlooking biomedical knowledge embedded in textual corpora that can provide complementary biological insights. Moreover, these methods inadequately integrate attribute-based representations (focusing on intrinsic properties) with structure-based representations (capturing heterogeneous relations) for modeling drug-cell interactions. To address these challenges, we propose a multi-modal contrastive learning method that integrates biological knowledge with dual-view molecular features. First, we incorporate textual descriptions of biological pathways and molecular mechanisms from authoritative databases as a complementary modality. Next, attribute-based representations are extracted from multi-omics profiles and drug molecular graphs, while structure-based representations are captured from drug-cell line interactions using a heterogeneous graph neural network. Finally, a contrastive learning module is designed to align these multi-modal representations through cross-modal learning (textual information vs. attribute view and intra-modal learning (attribute view vs. structure view), thereby establishing a unified semantic space for enhanced feature integration. Experiments conducted on the GDSC and CCLE datasets demonstrate that our proposed model outperforms previous state-of-the-art methods in DRP. Source code and datasets are available at <https://github.com/myduan18/CLMT>.

1. Introduction

Personalized cancer treatment has emerged as a transformative paradigm in modern oncology, with drug response prediction (DRP) serving as a fundamental component, but it remains challenging due to intrinsic cancer heterogeneity and individual variability (Wu et al., 2025d). Recent advances in pharmacogenomics (Pirmohamed, 2023; Sadee et al., 2023) have deepened our understanding of how genetic alterations and transcriptomic programming influence treatment outcomes, offering new opportunities for tailoring therapies to individual patients (Qahwaji et al., 2024). However, performing drug response testing on patients directly poses ethical challenges and incurs prohibitive costs. Consequently, cancer cell line data, which can be cultured on a large scale in vitro, has emerged as a viable alternative (Consortium, 2015). Moreover, with the advent of multi-omics sequencing technologies, cell lines can be profiled at the genomic, transcriptomic and proteomic levels, providing valuable and comprehensive molecular insights for predicting drug response (Hu et al., 2024; Jiang et al., 2025).

The increasing availability of high-throughput data has catalyzed the development of computational methods for accurate DRP. Deep learning approaches for DRP can be broadly categorized into three paradigms. (1) Attribute-based methods (Chen & Zhang, 2024; Hu et al., 2024; Li et al., 2024a) apply specialized feature extractors to independently learn molecular characteristics of cell lines and drugs before integrating them for prediction. (2) Structure-based methods (Nguyen et al., 2021; Zhu et al., 2022) prioritize modeling global-level associations among cell lines and drugs on biological network. (3) Hybrid methods have emerged (Wang et al., 2024; Xu et al., 2025; Yao & Zhang, 2024) to leverage the advantages of above two paradigms. This line of research can be further subdivided into two main strategies: cascaded strategy where molecular representations are first learned independently then used as node embeddings for relational learning (Wang et al., 2024), and joint learning strategy where attribute-based and structure-based learning are performed simultaneously and integrated through concatenation, distillation, or attention mechanisms (Xu et al., 2025, 2024; Yang & Li, 2023).

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Although existing methods demonstrate improvements in prediction performance, they still predominantly exploit attributive and structural features. While these features demonstrate strong descriptive power, they struggle to provide functional-level interpretations of biological mechanisms (Wu et al., 2025e; Yang & Li, 2023). Textual information exhibits unique advantages in this regard, functioning as a critical bridge linking molecular alterations to their functional implications (Luo et al., 2022; Pei et al., 2024). By contextualizing quantitative changes, such as high levels of gene expression, within biological processes like cell cycle dysregulation, textual data provide mechanistic explanations of molecular and cellular behavior, thereby facilitating knowledge integration and mitigating noise.

Recent work (Li et al., 2024) has shown the potential of natural language processing techniques in DRP contexts through label transformation, yet the rich textual descriptions of underlying molecular mechanisms remain largely unexploited as input features. Building on these observations, our motivation arises from the need to incorporate such biomedical knowledge embedded in texts into DRP. However, accomplishing this integration poses two fundamental challenges: (1) How to construct textual descriptions that effectively complement existing attributive and structural features with meaningful biological knowledge? (2) How to integrate these heterogeneous feature modalities to preserve their distinct informational contributions and maximize their synergistic potential for the DRP task?

To address the first challenge, we exploit textual information extracted from established biomedical databases including CCLE (Barretina et al., 2012), GeneCards (Safran et al., 2010) and DrugBank (Knox et al., 2024), encompassing cell line mutations, affected proteins, cellular pathways, and drug targets, to enrich domain-specific biological knowledge. Specifically, we characterize cell line functional states through pathway-level regulatory descriptions and establish standardized textual templates for both cell lines and drugs. These templates are then interconnected via shared or functionally related pathway functionalities, establishing biologically meaningful linkages between cellular states and drug mechanisms. A domain-specific language model, PubmedBert (Gu et al., 2021), which is pre-trained from scratch using abundant unlabeled biological texts, is subsequently fine-tuned to capture semantic insights from molecular mechanisms and pharmacological narratives.

As for the second challenge, we develop a multi-modal contrastive learning module that integrates heterogeneous features. This module employs intra-modal contrastive learning to unify molecular representations by aligning attribute and structure perspectives of the same biological entities. This approach maximizes the consistency between complementary views of identical cell lines and drugs while maintaining discrimination across different samples. Consequently, a unified embedding space is constructed that captures both detailed local interactions and broader system-level dependencies. Concurrently, cross-modal contrastive learning strategy addresses the semantic gap between textual descriptions and molecular features by aligning text-derived semantic representations with the attribute-based embeddings. This alignment ensures that the biological knowledge from textual descriptions complements the precise molecular information, thereby achieving synergistic multi-modal integration.

To this end, we propose a novel multi-modal contrastive learning method based on molecular data and biological text for DRP, namely CLMT. It consists of four components: textual feature representation, dual-view molecular feature representation, multi-modal contrastive learning and DRP predictor. Specifically, in the first module, we gather and format textual descriptions from authoritative biomedical databases. We then employ PubmedBert to encode this textual data, which encompasses molecular and pharmacological information to complement biological knowledge. The second module generates attribute view representations of cell lines and drugs, and utilizes heterogeneous graph convolution to model the associations among cell lines and drugs. Subsequently, the third module introduces multi-modal con-

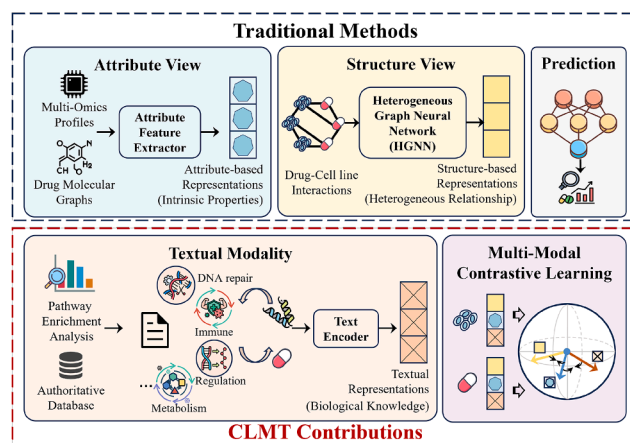


Fig. 1. Conceptual illustration of CLMT.

trastive learning to improve the consistency between the dual-view molecular and textual representations. Finally, the fourth module predicts the response value for each pair of cell line and drug.

In summary, Fig. 1 illustrates the high-level intuition behind CLMT's architecture design. The main contributions of this paper can be summarized as follows:

- We propose a multi-modal contrastive learning method for DRP by introducing pathway-level textual descriptions as input features to bridge the semantic gap between molecular measurements and functional mechanisms.
- We curate contextualized biomedical descriptions for drugs and cell lines, demonstrating that pathway-level textual representations enhance DRP performance.
- To integrate heterogeneous modalities, we design intra-modal and cross-modal contrastive learning strategies that utilize molecular attributes as an intermediate modality to align structural and textual representations in a shared embedding space.
- Extensive experiments demonstrate that CLMT consistently outperforms the state-of-the-art baselines on GDSC and CCLE datasets.

2. Related work

2.1. Multi-modal learning for DRP

Multi-modal learning integrates heterogeneous data to overcome the limitations of unimodal approaches. The effectiveness of integrating heterogeneous modalities has been demonstrated in various biomedical tasks. In medical informatics, researchers have enhanced clinical text classification by fusing contextual embeddings from language models with graph-structured medical knowledge (Wang et al., 2025b, 2023). In drug discovery, multi-modal approaches have successfully integrated pocket-drug graph structures with sequences information for accurate drug-target binding affinity prediction (Wang et al., 2025c) and unified heterogeneous relational views through contrastive learning for efficient drug repositioning (Wang et al., 2025).

Current deep learning-based DRP methods primarily treat multi-omics molecular profiles and drug chemical features (including SMILES strings and molecular graphs) as different modalities, employing specialized architectures such as graph neural networks for molecular graphs and neural encoders for omics vectors. These approaches consistently demonstrate superior performance compared to unimodal baselines by jointly modeling cellular and chemical contexts (Ge et al., 2025; Yu et al., 2025; Zheng et al., 2024). Beyond molecular features, recent advances have incorporated high-content imaging data and textual information to further enrich the multi-modal representations (Li et al., 2024,

2023). Nevertheless, there has been limited investigation into integrating multi-omics molecular characteristics with textual representations of functional pathways and pharmacological effects as input features.

Building upon previous multi-modal models, our method integrates more comprehensive modalities. For molecule features, we consider both attribute and structure views which capture intrinsic properties of individual entities and structural representations from global relationships. Furthermore, we incorporate textual modality, especially pathway-level enrichment description, to supplement molecular features with functional-level knowledge.

2.2. Contrastive learning for DRP

Contrastive learning has emerged as an effective paradigm for DRP, enabling the learning of discriminative representations through pulling positive samples closer while pushing negative samples apart. The application of contrastive learning in DRP can be summarized into two types: (1) Graph contrastive learning. This line models DRP as a bipartite or heterogeneous graph with drug and cell line nodes connected by response relationships or similar neighbors. These methods construct positive and negative samples through two strategies: directly using sensitive and resistant response labels directly (Liu et al., 2022; Yao & Zhang, 2024), or constructing different graph views of the same interactions as positive pairs (Dong et al., 2025). Contrastive learning is then applied at node-level or graph-level to enhance the quality of representations. (2) Multi-modal contrastive learning. These approaches align heterogeneous data types to exploit complementary knowledge across different modalities. For instance, MMCL-CDR (Li et al., 2023) incorporates omics and cell morphology images, while CLDR (Li et al., 2024) employ textual supervision, converting continuous response labels into natural language to guide representation learning.

In contrast to aforementioned contrastive learning models, CLMT explicitly considers semantic distances among modalities by designing two contrastive objectives which leverage attribute representations as bridges to align structural and functional features, thereby enabling progressive alignment toward a unified embedding space.

2.3. Pathway-informed methods for DRP

Biological pathway information is a critical component in enhancing DRP by providing mechanistic insights into cellular processes and drug mechanisms of action. The integration of pathway information in DRP can be categorized into three main strategies: (1) Pathway activity inference, which computes pathway enrichment scores or activity levels from gene expression data to capture dysregulated biological processes in cancer cell lines. Early work introduces kernelized Bayesian matrix factorization (KBMF), integrating pathway information as structured biological priors to learn drug response associations (Ammad-Ud-Din et al., 2016). Subsequent studies further utilize pathway knowledge by transforming it into quantitative features, such as pathway enrichment or activity scores for model input (Wang et al., 2019; Yang & Li, 2023). (2) Network-based propagation, such as SubCDR that employs subcomponent-guided method with pathway knowledge enhancement (Liu & Zhang, 2023). (3) Pathway-constrained architectures, where deep learning models incorporate pathway topology directly into their structure, such as through biologically informed neural networks (Ran et al., 2025; Snow et al., 2021; Zhang et al., 2021) and hierarchical attention mechanisms (Jin & Nam, 2021). These strategies demonstrate the value of pathway knowledge in improving the biological relevance of predictive models.

With the rise of large language models (LLMs), several DRP studies leverage the generative capabilities of LLMs for pathway-based feature selection (Carli et al., 2025) or serving as end-to-end DRP predictors via prompt engineering (Gnanaolivu et al., 2025). Our approach differs above methods by explicitly optimizing cross-modal alignment between molecular features and pathway-level text. This enables the proposed

model to capture fine-grained correspondences where molecular physicochemical patterns complement pathway functional mechanisms.

3. Task formulation

In this section, we first detail the collection and formulation of biological description templates for cell lines and drugs. We then define preliminary concepts and notations of molecular features and biological network construction. Finally, we formalize the DRP task.

3.1. Textual formation

This process consists of three steps: textual description collection, template construction, feature definition, and we will elaborate on each step in this section.

3.1.1. Textual description collection

We gather textual description from multiple authoritative databases to characterize both cell lines and drug.

Regarding each cell line, we extract descriptions across three aspects: foundational characteristics, pathway-level functional signatures, and genetic alterations.

- For foundational characteristics, we obtain cell line name, primary site, and histological classification from annotation files provided by the Cancer Cell Line Encyclopedia (CCLE) (Barretina et al., 2012) and Genomics of Drug Sensitivity in Cancer (GDSC) database (Yang et al., 2012).
- For pathway-level functional signatures, we employ single-sample Gene Set Enrichment Analysis (ssGSEA) to derive functional profiles (Barbie et al., 2009) using the 50 hallmark pathways from MSigDB (Liberzon et al., 2011) as reference gene sets. These hallmark pathways encompass key biological processes, such as DNA repair, metabolic processes, and immune responses, providing a comprehensive functional landscape. The top 3 ranked upregulation and downregulation pathway signatures are extracted. This pathway-based descriptions incorporates established biological knowledge, reduces noise inherent in high-dimensional genomic data, and improves information quality.
- For gene alteration, we identify functional alteration and mutated genes and query GeneCards (Safran et al., 2010) database to retrieve gene types, associated diseases, and pathway involvement. This provides precise functional characterization of the specific genetic alterations present in each cell line.

For each drug, we also extract descriptions across three key aspects: synonyms, molecular targets, and affected biological pathways.

- For drug synonyms, we obtain standardized drug names along with their alternative synonyms from annotation files provided by the GDSC database (Yang et al., 2012). This comprehensive naming information ensures accurate drug identification and facilitates cross-reference with other pharmaceutical databases and literature sources.
- For molecular targets, we select the primary targets for each compound, including specific proteins, enzymes, receptors, and other biomolecules that serve as direct interaction sites. This target information provides crucial mechanistic insights into the drug's mode of action and helps elucidate the molecular basis of therapeutic efficacy.
- For affected biological pathways, we identify the downstream functional alterations and cellular processes. These pathway annotations encompass various biological functions including signal transduction cascades, metabolic networks, cell cycle regulation, and apoptotic processes. Understanding pathway involvement is essential for predicting drug effects, potential side effects, and therapeutic applications.

[Cell line name] is a carcinoma cell line derived from [Primary site]. It is used in studies of [Histology] biology and therapeutic response. Pathway enrichment analysis reveals that [Cell line name] shows [Top Downregulated pathways] and [Top Upregulated pathways]. This cell line contains mutations in the [Mutation gene list] genes. [The function of each gene and its pathway].

[Drug name] is also known as [Synonyms]. [Drug Summary]. It is a pharmaceutical compound that targets [Target genes name]. These targets are involved in [Target pathway].

3.1.2. Template construction

To standardize the textual data collected for cell lines and drug compounds, we design structured templates to organize key information attributes. The design of the templates is guided by three key principles for effective textual representation of biological information. (1) Consistent ordering of attribute descriptions, enabling the model to focus on the essential differences inherent in biological semantics. (2) Strategic incorporation of transitional phrases to establish clear relationships between biological characteristics. (3) Hierarchical organization from basic properties to specific molecular mechanisms, making the generated text more intuitive to interpret. The textual descriptions establish semantic connections between cell lines and drugs through shared or functionally associated protein targets and signaling pathways. Specifically, we organize textual descriptions of cell lines and drugs according to the following templates:

3.1.3. Textual feature definition

Let $\mathcal{V}_{\text{CL}} = \{c_1, c_2, \dots, c_{|\text{CL}|}\}$ denote the set of cell lines and $\mathcal{V}_{\text{DR}} = \{d_1, d_2, \dots, d_{|\text{DR}|}\}$ denote the set of drugs, where $|\text{CL}|$ and $|\text{DR}|$ are the numbers of cell lines and drugs, respectively. Each cell line $c_i \in \mathcal{V}_{\text{CL}}$ is characterized by a textual description e_i^{CL} , encompassing tissue type, mutation genes and functional pathways. Similarly, each drug $d_j \in \mathcal{V}_{\text{DR}}$ is accompanied by textual input e_j^{DR} , such as pharmacodynamics and mechanism of action. We aim to learn representations of both cell lines and drugs from these textual descriptions through a domain-specific pre-trained language model.

3.2. Molecular feature definition

For each cell line c_i , there are multi-omics data including gene expression profiles $\mathbf{x}_i^{\text{ge}}$, somatic mutation profiles $\mathbf{x}_i^{\text{mut}}$, and copy number variation (CNV) profiles $\mathbf{x}_i^{\text{cnv}}$. Both $\mathbf{x}_i^{\text{mut}}$ and $\mathbf{x}_i^{\text{cnv}}$ are binary vectors: in $\mathbf{x}_i^{\text{mut}}$, 1 indicates gene mutation occurrence, while in $\mathbf{x}_i^{\text{cnv}}$, 1 represents amplification or deletion events. For each drug d_j , we obtain its Morgan fingerprint $\mathbf{f}_j$, a binary vector derived from the Simplified Molecular Input Line Entry System (SMILES) representation. Additionally, we convert the SMILES string of drug d_j into a 2D molecular graph using RDKit (Landrum & et al, 2013), denoted as $\mathcal{G}_j = \{\mathcal{V}_j, \mathcal{E}_j\}$, where $\mathcal{V}_j$ is the set of drug atoms and $\mathcal{E}_j$ is the set of chemical bonds.

3.3. Biological network construction

In this section, we define a heterogeneous biological network (HBN) that integrates three subnetworks, including the cell line similarity network, drug similarity network, and cell line-drug association network.

The cell line similarity network is constructed based on Pearson correlation coefficient of gene expression profiles, denoted as $\mathcal{G}_{\text{CL}} = \{\mathcal{V}_{\text{CL}}, \mathcal{E}_{\text{CL}}\}$, where $\mathcal{V}_{\text{CL}}$ and $\mathcal{E}_{\text{CL}}$ represent the node and edge sets of cell lines, respectively. Analogously, the similarity between drugs is calculated by drug fingerprints, the similarity network is represented as

$\mathcal{G}_{\text{DR}} = \{\mathcal{V}_{\text{DR}}, \mathcal{E}_{\text{DR}}\}$, where $\mathcal{V}_{\text{DR}}$ and $\mathcal{E}_{\text{DR}}$ represent the node and edge sets of drugs. The cell line-drug association network is an undirected graph, denoted as $\mathcal{G}_{\text{DRP}}$, the edges are constructed between drugs and cell lines if their $\ln(\text{IC50})$ measurements are available.

Subsequently, we define the HBN. Formally, given the entire node and edge sets, i.e., $\mathcal{V} = \mathcal{V}_{\text{CL}} \cup \mathcal{V}_{\text{DR}}$ and $\mathcal{E} = \mathcal{E}_{\text{CL}} \cup \mathcal{E}_{\text{DR}} \cup \mathcal{E}_{\text{DRP}}$, we design two type mapping functions, i.e., $\phi: \mathcal{V} \rightarrow \mathcal{T}^{\mathcal{V}}$ and $\psi: \mathcal{E} \rightarrow \mathcal{T}^{\mathcal{E}}$, to assign a specific type for each node and edge, where $\mathcal{T}^{\mathcal{V}}$ includes two node types, i.e., cell line and drug, $\mathcal{T}^{\mathcal{E}}$ encompasses three edge types, i.e., CL, DR and DRP. Hence, the HBN is eventually defined as $\mathcal{G}_{\text{HBN}} = \{\mathcal{V}, \mathcal{E}, \mathcal{T}^{\mathcal{V}}, \mathcal{T}^{\mathcal{E}}\}$.

3.4. DRP task

DRP is formulated as a supervised regression task, where our goal is to learn a function that maps the representation of a cell line-drug pair into a quantitative response value measuring the drug's efficacy against the specific cell line. In this task, we utilize the log-normalized half maximal inhibitory concentration ($\ln(\text{IC50})$) as the prediction target, where IC50 represents the drug concentration required to inhibit the growth or activity of a given cell line by 50%. A smaller $\ln(\text{IC50})$ value indicates higher cellular sensitivity to the drug, as it corresponds to a lower concentration required for 50% inhibition. Formally, given a cell line-drug pair (c_i, d_j) , we train a predictor model $\mathcal{M}(\cdot)$ to predict drug response value $\hat{y}_{i,j}$.

4. Methodology

As illustrated in Fig. 2, the overall architecture of CLMT comprises four modules, i.e., textual feature representation, dual-view molecular feature representation, multi-modal contrastive learning and DRP predictor. In the following subsections, we will elaborate on each component.

4.1. Textual feature representation

The standardized textual descriptions of cell line c_i and drug d_j are indicated as e_i^{CL} and e_j^{DR} , respectively. We leverage PubMedBERT to extract rich semantic representations from these structured texts. To better adapt the pre-trained language model to our task, we fine-tune its final linear layer to generate task-specific representations. Let $\Psi(\cdot)$ denote the textual feature extractor, and the textual representations t_i^{CL} and t_j^{DR} are obtained by $\Psi(e_i^{\text{CL}})$ and $\Psi(e_j^{\text{DR}})$.

4.2. Dual-view molecular feature representation

This section introduces two complementary views of molecular representation. The attribute view captures instance-level characteristics from cell line omics and drug molecular features, while the structure view models system-level interaction patterns through higher-order relationships.

4.2.1. Attribute representation of cell line

For each gene expression vector $\mathbf{x}_i^{\text{ge}}$ of cell line c_i , we utilize a multi layer perceptron (MLP) to map it into a low dimensional embedding $\mathbf{m}_i^{\text{ge}}$.

For each mutation vector $\mathbf{x}_i^{\text{mut}} \in \{0, 1\}^{d_n}$ of cell line c_i , we first embed each entry through a binary embedding matrix $\mathbf{E} \in \mathbb{R}^{2 \times d}$ (token embedding):

$$\mathbf{U}_i = \mathbf{E}[\mathbf{x}_i^{\text{mut}}], \quad (1)$$

where $\mathbf{U}_i \in \mathbb{R}^{d_n \times d}$ denotes the embedded mutation sequence matrix, with d_n and d representing the dimensions of the mutation data and the embedding space, respectively.

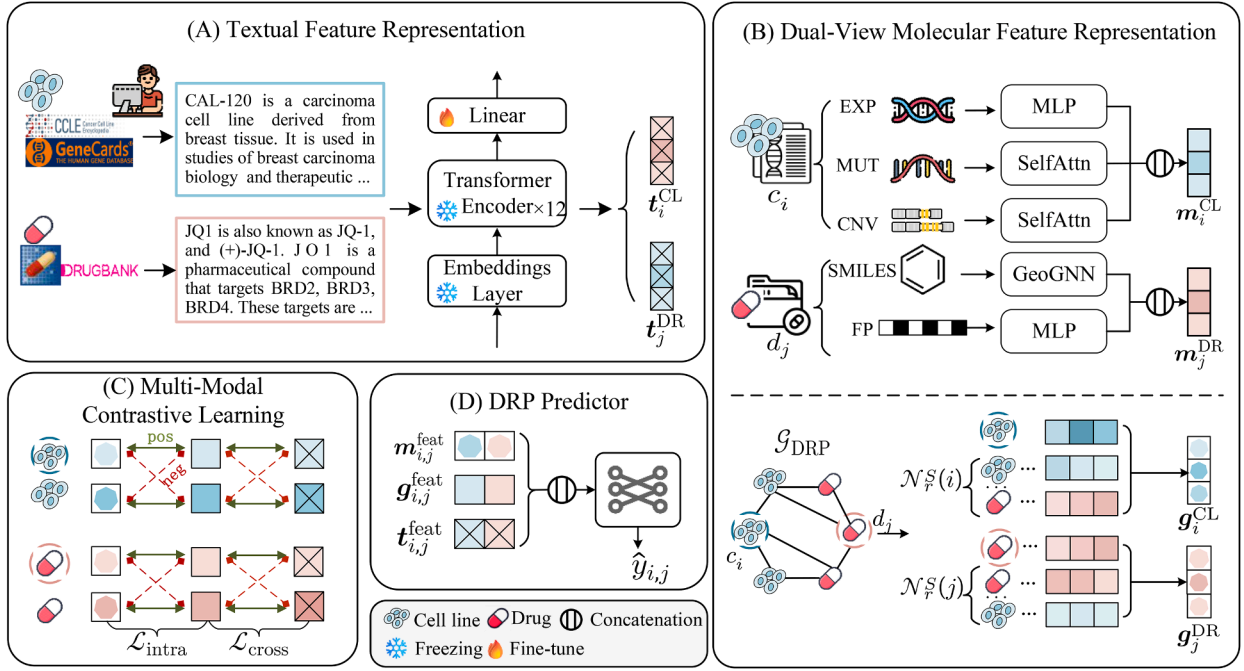


Fig. 2. The overall architecture of CLMT, which consists of four components, i.e., textual feature representation, dual-view molecular feature representation, multi-modal contrastive learning and DRP predictor.

We then apply a multi-head self-attention mechanism. For head $h = 1, \dots, H$:

$$Q_i^{(h)} = U_i W_Q^{(h)}, \quad K_i^{(h)} = U_i W_K^{(h)}, \quad V_i^{(h)} = U_i W_V^{(h)}, \quad (2)$$

where $W_Q^{(h)}, W_K^{(h)}, W_V^{(h)} \in \mathbb{R}^{d \times d_h}$ are query, key and value projection matrices, and $d_h = d/H$ is the head dimension, where d is the embedding dimension, H is the number of attention heads. The attention weights and values are computed as follows:

$$A_i^{(h)} = \text{softmax}\left(\frac{Q_i^{(h)}(K_i^{(h)})^T}{\sqrt{d_h}}\right), \quad H_i^{(h)} = A_i^{(h)}V_i^{(h)}, \quad (3)$$

where $A_i^{(h)} \in \mathbb{R}^{d_n \times d_n}$ denotes the normalized attention weights and $H_i^{(h)} \in \mathbb{R}^{d_n \times d_h}$ is the corresponding head output. The concatenated multi-head outputs O_i are then transformed through the output projection matrix W_O :

$$O_i = [H_i^{(1)} \parallel \dots \parallel H_i^{(H)}]W_O \quad (4)$$

A residual connection and layer normalization are then applied:

$$Z_i = \text{LN}_1(U_i + O_i) \quad (5)$$

Finally, a linear projection with W_{out} and bias b_{out} , followed by a second layer normalization LN_2 , yields the final mutation embedding m_i^{cnv} :

$$m_i^{mut} = \text{LN}_2(W_{out}\text{MeanPool}(Z_i) + b_{out}) \quad (6)$$

where $\text{MeanPool}(\cdot)$ is the mean pooling function. Similarly, the mutation embedding m_i^{cnv} is also obtained using multi-head self-attention mechanism. Finally, the cell line representation m_i^{CL} of cell line c_i is formed by concatenating three omics embeddings: $m_i^{CL} = m_i^{se} \parallel m_i^{mut} \parallel m_i^{cnv}$.

4.2.2. Attribute representation of drug

We encode the molecular graph G_j of drug d_j using a two-stage approach. First, we apply Graph Isomorphism Network (GIN) (Xu et al., 2018) to update atom representations by aggregating information from neighboring atoms with bond feature embeddings. Specifically, the l -th

layer GIN update with bond information is defined as:

$$m_{j,u}^{(l)} = \text{MLP}^{(l)}\left(m_{j,u}^{(l-1)} + \sum_{v \in \mathcal{N}(u)} \sigma(m_{j,v}^{(l-1)} + d_{uv}^b)\right) \quad (7)$$

where $m_{j,u}^{(l)}$ denotes the representation for atom u of drug j at the l -th layer, $\sigma(\cdot)$ is the ReLU activation function, d_{uv}^b is the initial bond attribute vector for bond uv , $\mathcal{N}_j(u)$ is the set of neighboring atoms of atom u , and $m_{j,u}^{(0)} = d_{uv}^b$.

Subsequently, we employ Geometry-enhanced Graph Neural Network (GeoGNN) (Fang et al., 2022) to incorporate 3D geometric information by refining bond representations with angle information and updating atom features accordingly. The bond uv 's representation $m_{j,uv}^{(l)}$ of drug d_j at the l -th layer is updated by:

$$m_{j,uv}^{(l)} = \text{MLP}^{(l)}\left(m_{j,uv}^{(l-1)} + \sum_{w \in \mathcal{N}(v) \setminus u} \sigma(m_{j,vw}^{(l-1)} + d_{uvw}^a) + \sum_{w \in \mathcal{N}(u) \setminus v} \sigma(m_{j,uw}^{(l-1)} + d_{uwv}^a)\right) \quad (8)$$

where d_{uvw}^a is the angle attribute vector for the angle formed by atoms $u-v-w$, and $m_{j,uv}^{(0)} = d_{uv}^b$. The bond representation is updated at layer k , $m_{j,u}^{(k)}$ is reformulated by:

$$m_{j,u}^{(l)} = \text{MLP}^{(l)}\left(m_{j,u}^{(l-1)} + \sum_{v \in \mathcal{N}(u)} \sigma(m_{j,v}^{(l-1)} + m_{j,uv}^{(l-1)})\right) \quad (9)$$

After updating the atom and bond representations through GIN and GeoGNN layers, for each drug d_j , we aggregate the learned atomic features using a max pooling operation across all atom nodes of drug to obtain the structure view representation m_j^{mol} . Furthermore, we utilize MLP to capture the features of fingerprint f_j , obtaining the representation m_j^{fp} . Finally, representation m_j^{DR} of drug d_j is formed by concatenating two representations: $m_j^{DR} = m_j^{mol} \parallel m_j^{fp}$.

4.2.3. Structural feature representation

To effectively capture the diverse node types and edge relationships in biological network G_{HBN} , we develop a customized heterogeneous GraphSAGE (Hamilton et al., 2017) model. GraphSAGE relies on the

neighborhood aggregation principle that a node's representation can be learned by aggregating features from its local neighbors. Importantly, our network construction inherently satisfies a local smoothness property by connecting nodes with high molecular similarity. This introduces local consistency in node features and makes neighborhood aggregation well aligned with the underlying biological relationships. For each layer l , embedding update process of node v is divided into the following steps:

Step 1: type-specific projection. For each node v of type $\phi(v)$, we employ type-specific projection matrix to transform node features:

$$\mathbf{h}_v^{(l)} = \mathbf{W}_{\phi(v)}^{(l)} \mathbf{h}_v^{(l-1)} + \mathbf{b}_{\phi(v)}^{(l)} \quad (10)$$

where $\mathbf{h}_v^{(l)}$ and $\mathbf{h}_v^{(l-1)}$ denote the embedding vectors of node v at layer l and $l-1$, $\mathbf{W}_{\phi(v)}^{(l)}$ represents the node type-specific parameter matrix, and $\mathbf{b}_{\phi(v)}^{(l)}$ is a bias vector.

Step 2: type-specific neighborhood sampling. For each edge type $r \in \mathcal{T}^E$, a size- K type-specific neighborhood $\mathcal{N}_r^K(v)$ is sampled from all neighbors.

Step 3: node type-aware message passing and aggregation. For each edge type r , we first implement edge-type specific mean aggregation to obtain the neighborhood embedding $\mathbf{h}_{\mathcal{N}(r)}^{(l)}$:

$$\mathbf{h}_{\mathcal{N}(r)}^{(l)} = \frac{1}{|\mathcal{N}_r^K(v)|} \sum_{u \in \mathcal{N}_r^K(v)} \mathbf{h}_u^{(l)} \quad (11)$$

we then fuse information from node itself and its neighborhood:

$$\mathbf{z}_r^{(l)}(v) = \mathbf{W}_{1,r}^{(l)} \mathbf{h}_v^{(l)} + \mathbf{W}_{2,r}^{(l)} \mathbf{h}_{\mathcal{N}(r)}^{(l)} \quad (12)$$

where $\mathbf{W}_{1,r}$ and $\mathbf{W}_{2,r}$ are edge-type specific transformation matrices that control the importance of self-information and neighborhood information, respectively.

Step 4: node representation updating. The embedding of node v at l -th layer is updated by aggregating messages from all types edges:

$$\mathbf{g}_v^{(l)} = \sigma\left(\frac{1}{|\mathcal{T}^E|} \sum_{r \in \mathcal{T}^E} \alpha_r^{(l)} \mathbf{z}_r^{(l)}(v)\right) \quad (13)$$

where $\alpha_r^{(l)}$ denotes the learnable weight for edge type r at layer l . After stacking L layers, the final node representations effectively encode both the features of each node and the complex topological structures surrounding them. Specifically, the obtained representations of cell lines and drugs from $\mathcal{G}_{\text{HBN}}$ at L -th layer are denoted as $\{\mathbf{g}_i^{\text{CL}}\}_{i=1}^{|\mathcal{V}^{\text{CL}}|}$ and $\{\mathbf{g}_j^{\text{DR}}\}_{j=1}^{|\mathcal{V}^{\text{DR}}|}$, respectively (superscript L is omitted).

4.3. Multi-modal contrastive learning

At this stage, we have obtained representations from the textual and dual-view molecular feature modules. For cell line c_i , these are $\mathbf{t}_i^{\text{CL}}$, $\mathbf{m}_i^{\text{CL}}$, and $\mathbf{g}_i^{\text{CL}}$; for drug d_j , these are $\mathbf{t}_j^{\text{DR}}$, $\mathbf{m}_j^{\text{DR}}$, and $\mathbf{g}_j^{\text{DR}}$. We formalize the semantic gap between molecular and textual modalities through the latent class framework of [Saunshi et al. \(2019\)](#), [Tosh et al. \(2021\)](#). In our setting, each drug or cell line pair is sampled from the distribution of a latent class representing the underlying biological concept (e.g., “epidermal growth factor receptor inhibition in non-small cell lung cancer”). The molecular features and textual descriptions are different observations of the same latent class but occupy distinct feature spaces, leading to the semantic gap.

To bridge this semantic gap, we design the multi-modal contrastive learning module. Considering the varying semantic proximity across modalities, attribute representations are utilized as an intermediary to connect structural and textual features, thereby narrowing the semantic gap between molecular data and biological text. Specifically, it comprises two strategies: intra-modal contrastive learning and cross-modal contrastive learning.

The intra-modal contrastive learning aims to align vector representations of the same entity (cell line or drug) extracted from attribute view

and structure view. $\mathbf{m}_i^{\text{CL}}$ and $\mathbf{g}_i^{\text{CL}}$ corresponding to the cell line c_i are treated as a positive pair, while representations from different cell lines within the datasets form negative pairs. A similar strategy is adopted for drug representations $\mathbf{m}_j^{\text{DR}}$ and $\mathbf{g}_j^{\text{DR}}$. For each cell line-drug pair of (c_i, d_j) , the intra-modal contrastive loss is applied separately to each cell line and each drug, thereby minimizing the embedding distance between positive pairs while maximizes the distance between negative pairs.

We adopt the InfoNCE ([Chen et al., 2020](#)) to formulate this objective, which assumes that negative samples are independently sampled from the data distribution to provide an unbiased mutual information estimate. In practice, we approximate the i.i.d. assumption by sampling negative pairs from distinct biological entities within each batch. It can be formulated as follows:

$$\begin{aligned} \mathcal{L}_{\text{intra}} = & - \sum_{c_i \in \mathcal{V}^{\text{CL}}} \log \frac{\exp(f(\mathbf{m}_i^{\text{CL}}, \mathbf{g}_i^{\text{CL}})/\tau)}{\sum_{c_u \in \mathcal{V}^{\text{CL}}} \exp(f(\mathbf{m}_i^{\text{CL}}, \mathbf{g}_u^{\text{CL}})/\tau)} \\ & - \sum_{d_j \in \mathcal{V}^{\text{DR}}} \log \frac{\exp(f(\mathbf{m}_j^{\text{DR}}, \mathbf{g}_j^{\text{DR}})/\tau)}{\sum_{d_v \in \mathcal{V}^{\text{DR}}} \exp(f(\mathbf{m}_j^{\text{DR}}, \mathbf{g}_v^{\text{DR}})/\tau)} \end{aligned} \quad (14)$$

where τ is a temperature coefficient.

The cross-modal contrastive learning is performed at the instance level, aligning molecular representations with their corresponding textual representations for each cell line and each drug. For cell line c_i , we treat $\mathbf{m}_i^{\text{CL}}$ and $\mathbf{t}_i^{\text{CL}}$ as a positive pair, while representations from different cell lines form negative pairs. Drug representations $\mathbf{m}_j^{\text{DR}}$ and $\mathbf{t}_j^{\text{DR}}$ of d_j are organized in the same strategy. The cross-modal contrastive loss is applied to minimize the embedding distance between positive pairs while maximizing the distance between negative pairs:

$$\begin{aligned} \mathcal{L}_{\text{cross}} = & - \sum_{c_i \in \mathcal{V}^{\text{CL}}} \log \frac{\exp(f(\mathbf{m}_i^{\text{CL}}, \mathbf{t}_i^{\text{CL}})/\tau)}{\sum_{c_u \in \mathcal{V}^{\text{CL}}} \exp(f(\mathbf{m}_i^{\text{CL}}, \mathbf{t}_u^{\text{CL}})/\tau)} \\ & - \sum_{d_j \in \mathcal{V}^{\text{DR}}} \log \frac{\exp(f(\mathbf{m}_j^{\text{DR}}, \mathbf{t}_j^{\text{DR}})/\tau)}{\sum_{d_v \in \mathcal{V}^{\text{DR}}} \exp(f(\mathbf{m}_j^{\text{DR}}, \mathbf{t}_v^{\text{DR}})/\tau)} \end{aligned} \quad (15)$$

The total loss of the multi-modal contrastive learning module is formulated as:

$$\mathcal{L}_{\text{C}} = \mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{cross}} \quad (16)$$

4.4. DRP predictor

Given each cell line-drug pair (c_i, d_j) , the DRP predictor aims to predict the $\ln(\text{IC}_{50})$ value, which reflect the sensitivity of cells to drugs. Specifically, we concatenate these embeddings to form cell line-drug pair representations, including attribute view $\mathbf{m}_{i,j}^{\text{feat}} = \mathbf{m}_i^{\text{CL}} || \mathbf{m}_j^{\text{DR}}$, structure view $\mathbf{g}_{i,j}^{\text{graph}} = \mathbf{g}_i^{\text{CL}} || \mathbf{g}_j^{\text{DR}}$, and textual representations $\mathbf{t}_{i,j}^{\text{text}} = \mathbf{t}_i^{\text{CL}} || \mathbf{t}_j^{\text{DR}}$. They are fed into MLP to predict response value:

$$\hat{y}_{i,j} = \text{MLP}(\mathbf{m}_{i,j}^{\text{feat}} || \mathbf{g}_{i,j}^{\text{graph}} || \mathbf{t}_{i,j}^{\text{text}}) \quad (17)$$

Mean squared error is utilized as the loss function for model optimization:

$$\mathcal{L}_{\text{DRP}} = \frac{1}{N} \sum_{(i,j) \in \Omega} (\hat{y}_{i,j} - y_{i,j})^2 \quad (18)$$

where Ω denotes the set of all sample indices, $y_{i,j}$ is the ground-truth label and N is dataset size.

Additionally, following [Chen and Zhang \(2024\)](#), we introduce a regularization term $\mathcal{L}_{\text{R}}$ that incorporates three biological constraints: cancer types, drug targets pathways and drug sensitivity threshold derived from large-scale pharmacogenomic studies ([Iorio et al., 2016](#)). These biological constraints encourage the model to learn representations that are consistent with known pharmacological principles, thereby improving model generalization. From statistical learning theory, the generalization bound is characterized by $\mathcal{R}(f_{\theta}) \leq \hat{\mathcal{R}}(f_{\theta}) + \mathcal{R}_n(\mathcal{H})$, where f_{θ} denotes the prediction model parameterized by θ , $\mathcal{R}(f_{\theta})$ denotes the

Table 1
Statistic of Two Datasets.

Datasets	Sample Number			Feature Number		
	Cell Line	Drug	Response	Gene Expression	Mutation	CNV
GDSC	734	177	125,696	8000	636	694
CCLL	385	23	8673	8000	444	700

expected risk over the true data distribution, $\hat{R}(f_\theta)$ is the empirical risk on the training set, and $\mathcal{H}$ denotes the hypothesis space, $\mathcal{R}_n(\mathcal{H})$ denotes the Rademacher complexity (Bartlett & Mendelson, 2002), which measures the $\mathcal{H}$'s capacity to fit random noise and directly controls the generalization gap between empirical and expected risks. The biological regularization term $\mathcal{L}_R$ constrains the hypothesis space $\mathcal{H}$ by penalizing predictions that violate known biological relationships, effectively restricting learning to a structured subspace consistent with biological priors. This constraint reduces the effective hypothesis complexity, thereby tightening the generalization bound and enhancing robustness to out-of-distribution samples.

Finally, the total loss function, denoted as $\mathcal{L}_{\text{Total}}$, consists of the main DRP prediction loss, multi-modal contrastive learning loss, and regularization term:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{DRP}} + \alpha \mathcal{L}_C + \beta \mathcal{L}_R \quad (19)$$

where α and β are hyperparameters to balance contrastive learning loss and regularization term. They are jointly optimized by multi-task learning. The trade-off between data-driven learning and knowledge-guided regularization is controlled by the hyperparameter β . When $\beta \rightarrow 0$, the model relies purely on empirical risk minimization, increasing its proneness to overfitting spurious correlations. Conversely, large β values inject strong biological priors as inductive bias, guiding the model toward biologically plausible solutions particularly when training data insufficiently covers the mechanism space.

5. Experiments

5.1. Materials

We evaluate our model performance using two commonly-used datasets: CCLL (Barretina et al., 2012) and GDSC (Yang et al., 2012) datasets.

Following the data preprocessing protocol established by Chen and Zhang (2021, 2024), for each dataset, we obtain three types of omics data. Specifically, we retain 8000 genes with the highest expression variance across samples. For mutation and CNV data, we apply gene filtering based on cancer-specific genes cataloged in the COSMIC database. Table 1 lists the detailed statistics of two datasets.

5.2. Experimental setup

5.2.1. Evaluation metrics

Four evaluation metrics are chosen to evaluate prediction performance, including root mean square deviation (RMSE), mean absolute error (MAE), Pearson Correlation Coefficient (PCC) and R-squared (R^2). For the regression error metrics RMSE and MAE, lower values correspond to better performance, while for the correlation evaluation metrics PCC and R^2 , higher scores indicate better performance.

5.2.2. Implementation details

Following previous work (Chen & Zhang, 2024), we divide the dataset into 80% development set and 20% test set. The development set is subsequently partitioned into 90% training and 10% validation subsets. We employ early stopping during training and evaluate the final model on the test set. To alleviate experimental bias, we repeat this entire procedure 5 times and report the averaged results.

Table 2
Text Length Distributions of Two Datasets.

Datasets	Components	Cell line	
		Average Tokens	Median Tokens
GDSC	Basic Info	31.70	30.00
	Pathway	94.69	95.00
	Mutation	2270.95	1519.00
	All Text	2393.34	1641.50
CCLL	Basic Info	37.35	35.00
	Pathway	98.06	97.00
	Mutation	1299.63	842.00
	All Text	1431.23	975.00
Datasets	Components	Drug	
		Average Tokens	Median Tokens
GDSC	All Text	97.25	51.00
CCLL	All Text	129.70	90.00

The embedding dimension of attribute, structure, and textual representations is set to 256, the number of neighbors to construct similar network is 20, and the number of graph neural network layers is 2. Dropout rate of 0.3 is applied after each hidden layer to prevent overfitting. The batch size is set to 256. The contrastive learning coefficient α is tuned among {0.001, 0.005, 0.01, 0.015} and the regularization coefficient β is tuned among {0.1, 0.3, 0.5, 0.7}. Moreover, as for temperature coefficient τ , we tune it among {0.1, 0.3, 0.5, 0.7}. We apply Adam-based optimization with an initial learning rate of 1e-3. A Multi-StepLR scheduler is applied to decay the learning rate by 0.1.

All texts are truncated to a maximum length of 512 tokens due to the input limitation of the PubMedBERT model. We perform the statistical analysis of text length in our datasets utilizing PubMedBERT tokenizer. The results are presented in Table 2. For cell lines, basic information and pathway components are more concise, typically comprising fewer than 100 tokens. In contrast, mutated genes are characterized by significantly more extensive descriptive annotations. For drug descriptions, they exhibit moderate length, with average token counts of 97.25 and 129.70 for the two datasets, respectively. As discussed in Section 3.1.2, we employ a hierarchical truncation strategy. Specifically, drug descriptions are fully retained, as their lengths remain below this token limitation. While for cell lines, when truncation is necessary, we prioritize preserving the global biological descriptions, including basic information as well as pathway-level text, and the gene list derived from mutation text. This design ensures that the essential molecular context relevant to DRP is preserved, despite partial loss of detailed gene-level textual annotations.

5.2.3. Baselines

The baselines are categorized into the following five groups: (1) Matrix factorization method: SRMF (Wang et al., 2017) (2) Attribute-based methods: DeepCDR (Liu et al., 2020), NeRD (Cheng et al., 2022), Hi-GeoMVP (Chen & Zhang, 2024) (3) Structure-based methods: GraphDRP (Nguyen et al., 2021), TGSA (Zhu et al., 2022) (4) Hybrid methods: Multi-DRP (Wang et al., 2024), RedCDR (Xu et al., 2024) (5) Text-based method: CLDR (Li et al., 2024).

To ensure fair comparison across all methods, all baselines are evaluated using identical experimental setting, including the same data splits, omics features, and response labels.

5.3. Main results and analyses

As shown in Table 3, we present the performance comparison results of CLMT and competitive baselines on the GDSC and CCLL datasets. The results are reported as (mean $\pm$ standard deviation). The best score is highlighted in **bold**, the second best score is underlined. “*” and “***” indicate that our proposed method significantly outperforms the

Table 3
Performance Comparison on Two Datasets.

Datasets	Methods	Evaluation Criteria			
		RMSE↓	MAE↓	PCC↑	R2
GDSC	SRMF	1.116 ± 0.0326	0.803 ± 0.0232	0.913 ± 0.0041	0.830 ± 0.0035
	DeepCDR	1.110 ± 0.0298	0.864 ± 0.0309	0.926 ± 0.0026	0.832 ± 0.0090
	GraphDRP	1.102 ± 0.0134	0.853 ± 0.0118	0.924 ± 0.0011	0.834 ± 0.0041
	TGSA	1.009 ± 0.0053	0.744 ± 0.0053	0.928 ± 0.0008	0.861 ± 0.0015
	NeRD	0.969 ± 0.0080	0.705 ± 0.0056	0.934 ± 0.0012	0.872 ± 0.0021
	MultiDRP	1.001 ± 0.0391	0.721 ± 0.0053	0.930 ± 0.0035	0.860 ± 0.0080
	RedCDR	0.929 ± 0.0022	0.684 ± 0.0018	0.939 ± 0.0011	0.873 ± 0.0017
	Hi-GeoMVP	0.931 ± 0.0030	0.680 ± 0.0018	0.941 ± 0.0006	0.880 ± 0.0008
	CLDR	0.929 ± 0.0024	0.678 ± 0.0021	0.942 ± 0.0009	0.882 ± 0.0011
	CLMT (ours)	0.924 ± 0.0030*	0.674 ± 0.0033	0.945 ± 0.0018*	0.888 ± 0.0006**
CCLE	SRMF	1.209 ± 0.0113	0.704 ± 0.0091	0.824 ± 0.0039	0.654 ± 0.0058
	DeepCDR	1.156 ± 0.0102	0.675 ± 0.0110	0.828 ± 0.0052	0.676 ± 0.0043
	GraphDRP	1.097 ± 0.0097	0.663 ± 0.0116	0.832 ± 0.0042	0.688 ± 0.0054
	TGSA	1.032 ± 0.0102	0.653 ± 0.0120	0.844 ± 0.0024	0.713 ± 0.0022
	NeRD	1.111 ± 0.0182	0.654 ± 0.0172	0.842 ± 0.0049	0.686 ± 0.0103
	MultiDRP	1.042 ± 0.0133	0.649 ± 0.0123	0.845 ± 0.0040	0.704 ± 0.0076
	RedCDR	1.095 ± 0.0107	0.655 ± 0.0094	0.850 ± 0.0009	0.711 ± 0.0015
	Hi-GeoMVP	0.992 ± 0.0118	0.653 ± 0.0112	0.855 ± 0.0012	0.695 ± 0.0034
	CLDR	0.987 ± 0.0098	0.650 ± 0.0082	0.867 ± 0.0022	0.744 ± 0.0023
	CLMT (ours)	0.974 ± 0.0093	0.635 ± 0.0146*	0.886 ± 0.0011**	0.771 ± 0.0041**

strongest baseline on the corresponding metric, as assessed by paired *t*-test ($p < .05$ and $p < .01$, respectively). From the results, we observe that CLMT outperforms all competitive baselines on both datasets, indicating that our proposed method has superior predictive ability. CLMT also presents statistically significant improvements in terms of most evaluation metrics with paired *t*-test. Furthermore, from the experimental results, we also make the following observations.

Traditional SRMF exhibits relatively weak performance. This indicates inherent limitations of simple matrix factorization techniques in capturing complex interactions between drugs and cell lines.

Attribute-based methods demonstrate superior overall performance, with Hi-GeoMVP emerging as the strongest baseline. This line of research tends to achieve better results when equipped with more sophisticated feature extractors, such as considering the gene-gene interactions. In particular, compared to DeepCDR and NeRD, Hi-GeoMVP achieves better performance by ensembling multiple types of feature extractors to capture both omics and drug characteristics. However, despite their strong predictive capabilities, attribute-based methods primarily rely on vectorized omics profiles such as gene expression or mutation features and often overlook the intrinsic relational structures among biological entities. To address this limitation, our approach incorporates structure view representations to model global topological relationships. Moreover, intra-modal contrastive objective effectively fuses these two perspectives, allowing the model to leverage both fine-grained feature information and graph structure in a unified embedding space.

Among structure-based approaches, TGSA achieves competitive performance, particularly on the CCLE dataset, which can be attributed to its utilization of pre-trained chemical embeddings for drugs. However, these structure-based methods primarily only focus on structural relationships and overlook valuable instance-level information. In contrast, CLMT introduces an attribute-based perspective that provides complementary, instance-level details. Furthermore, the contrastive learning promotes more robust and generalizable representations by bridging the gap between structural and feature-level information.

CLMT consistently prevails over hybrid-based baselines. We attribute this improvement to the integration of biological textual information, which fundamentally distinguishes our method from unimodal omics models. Through cross-modal contrastive learning, our model aligns molecular representations with semantically rich descriptions (e.g., molecular mechanism, drug target pathways), thereby injecting domain

knowledge that absent from the molecular data, enabling biologically meaningful representations for better DRP performance.

We further compare CLMT with CLDR to validate the effectiveness of incorporating biological textual knowledge. CLDR is a contrastive learning framework that aligns extracted drug and cell line features with label text descriptions. For fair comparison, we implement CLDR with the Hi-GeoMVP encoder. As shown in Table 3, CLMT demonstrates superior performance by explicitly integrating rich biological mechanism knowledge from curated databases through multi-modal contrastive learning. Unlike CLDR, which relies solely on label text for representation alignment, CLMT learns pharmacologically grounded representations that enable more robust and biologically meaningful predictions.

5.4. Ablation study

To evaluate the contribution of each component, we conduct comprehensive ablation studies across GDSC and CCLE datasets. The ablation results are shown in Table 4.

First, to quantify the contribution of each text component, we conduct ablation experiments by removing basic, pathway, and mutation text individually. As shown in Table 4, pathway text removal causes the most significant performance degradation across both datasets, followed by basic text and mutation text. These results confirm that pathway-level knowledge provides the most critical mechanistic understanding for DRP, without which effective cross-modal alignment in contrastive learning is hindered.

Subsequently, when the attribute view representations derived from molecular data are removed, i.e. w/o attribute view, the ablation model shows worse performance, indicating that molecular characteristics constitute a critical component for DRP. Removing structure representations also leads to performance degradation, demonstrating that these modalities provide complementary semantic information crucial for improving DRP accuracy.

Next, we remove the contrastive learning module, the corresponding ablation performance decrease highlights the crucial role of aligning dual-level features from omics data and biological text. This module enables the mapping of different modality features into a unified semantic space, thereby enhancing the quality and consistency of learned representations.

Table 4
Ablation Studies on Two Datasets.

Datasets	Methods	Evaluation Criteria			
		RMSE↓	MAE↓	PCC↑	R ² ↑
GDSC	CLMT	0.924	0.674	0.945	0.889
	w/o Basic Text	0.928	0.678	0.942	0.884
	w/o Pathway Text	0.932	0.682	0.941	0.880
	w/o Mutation Text	0.927	0.677	0.943	0.887
	w/o Attribute View	0.957	0.712	0.935	0.864
	w/o Structure View	0.928	0.687	0.940	0.885
	w/o Contrastive Learning	0.951	0.697	0.937	0.878
	w/o Regularization	0.938	0.685	0.940	0.884
	-Three Pairwise CTL	0.928	0.680	0.940	0.883
CCLE	-Triplet Attention	0.930	0.681	0.941	0.881
	CLMT	0.974	0.605	0.886	0.771
	w/o Basic Text	0.978	0.615	0.884	0.768
	w/o Pathway Text	0.983	0.622	0.880	0.762
	w/o Mutation Text	0.976	0.611	0.883	0.768
	w/o Attribute View	1.032	0.716	0.865	0.729
	w/o Structure View	0.976	0.642	0.881	0.760
	w/o Contrastive Learning	0.996	0.726	0.871	0.746
	w/o Regularization	0.984	0.614	0.882	0.767
	-Three Pairwise CTL	0.981	0.647	0.872	0.756
	-Triplet Attention	1.011	0.654	0.866	0.731

Moreover, we ignore the regularization item. This ablation variant shows worse performance, demonstrating that the regularization item appears to enforce consistency with known biological principles, thereby improving model generalization in DRP tasks.

Finally, to further demonstrate the effectiveness of the proposed multi-modal contrastive learning module, we replace it with two alternative fusion strategies: (1) three pairwise contrastive learning objectives (denoted as three pairwise CTL), which extend the CLMT by incorporating structure-text contrastive learning, and (2) triplet attention, which captures pairwise interactions among all three modalities via shared attention mappings (Misra et al., 2021). As shown in Table 4, CLMT consistently outperforms these alternatives on both the GDSC and CCLE datasets. These results indicate that neither directly adding cross-modal contrastive pairs nor adopting triplet attention effectively addresses the semantic heterogeneity across modalities, thereby leading to performance degradation. In contrast, the attribute view serves as an intermediate bridge, enabling a progressive alignment strategy where each step involves manageable semantic transitions, thus proving crucial for effective multi-modal learning.

5.5. Hyperparameter sensitivity analysis

The sensitivity analysis results of contrastive learning coefficient α and regularization coefficient β are shown in Fig. 3. A smaller value of α results in inefficient alignment and loss of complementarity, while a larger value impedes discrimination ability of the model. For coefficient β , CLMT gets the best performance at $\beta = 0.5$ on both GDSC and CCLE. A larger β value tends to constrain the model's learning capacity, resulting in underfitting and performance degradation. However, the overall performance fluctuations are relatively small across a reasonable range of α and β , indicating that our model is robust to the choice of these hyperparameters.

We further investigate the role of temperature τ in the Eq. 14–15. Smaller values of τ sharpen the similarity logits, enforcing stronger alignment between positive pairs but at the risk of collapsing local neighborhoods, whereas larger values soften the contrastive objective, leading to a more uniformly distributed embedding space but weaker discriminative power. Both datasets reach optimal RMSE performance at $\tau = 0.3$, demonstrating that moderate temperature balances discriminative alignment with representational flexibility. Through the InfoNCE

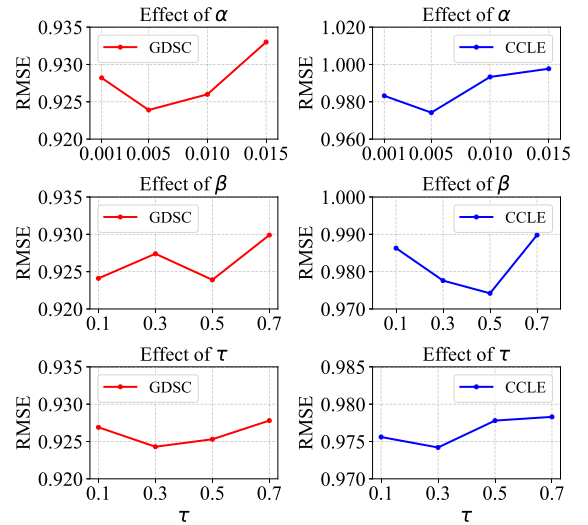


Fig. 3. Hyperparameter sensitivity analysis of α , β and τ .

objective, contrastive learning encourages semantic representations that capture latent biological factors invariant across heterogeneous views while suppressing modality-specific noise, thereby promoting generalization robustness to domain shifts inherent in biological data.

5.6. Cell line/drug-blind test

To further evaluate the robustness of our model and its potential in precision medicine. We conduct cell line-blind and drug-blind tests to simulate real-world scenarios: predicting the response value of existing drugs on previously unseen cancer cell lines and evaluating the efficacy of newly discovered drugs on known cancer cell lines. In the cell line-blind test, we divide the cell lines into training, validation, and test sets based on their names. For the drug-blind test, drugs are split based on their scaffold structures (Hu et al., 2019), which provide a more realistic estimation of model performance. The results are shown in Fig. 4.

Compared to the cell-blind test, the drug-blind test is more challenging due to the expansive and diverse chemical space of drugs, which limits the model's ability to learn transferable representations from a small set of screened drugs (Partin et al., 2023). From the results, CLMT outperforms all the baselines in predicting drug responses for new cell lines and compounds. CLDR shows modest gains compared to Hi-GeoMVP, particularly in the cell line-blind scenario. The superior performance of CLMT results from the additional incorporation of textual features and multi-modal contrastive learning strategy. Unlike purely attribute-based and structure-based approaches, textual descriptions provide semantic context about drug mechanisms of action, enabling the model to learn generalizable biological principles. The multi-modal contrastive learning strategy aligns features from different modalities of identical cell lines and drugs, thereby enhancing semantic consistency and representation quality.

5.7. The joint optimization of contrastive learning

Under the framework (Tosh et al., 2021), positive pairs in contrastive learning correspond to different views generated from the same underlying latent semantic variable, while negative samples are drawn from different latent variable realizations. Minimizing the contrastive objective facilitates feature representations that share common latent semantics to concentrate in the embedding space, while suppressing model-specific noise. Our design is theoretically justified because all three representations originate from the same biological entity and therefore share underlying biological semantics, enabling contrastive learning to recover

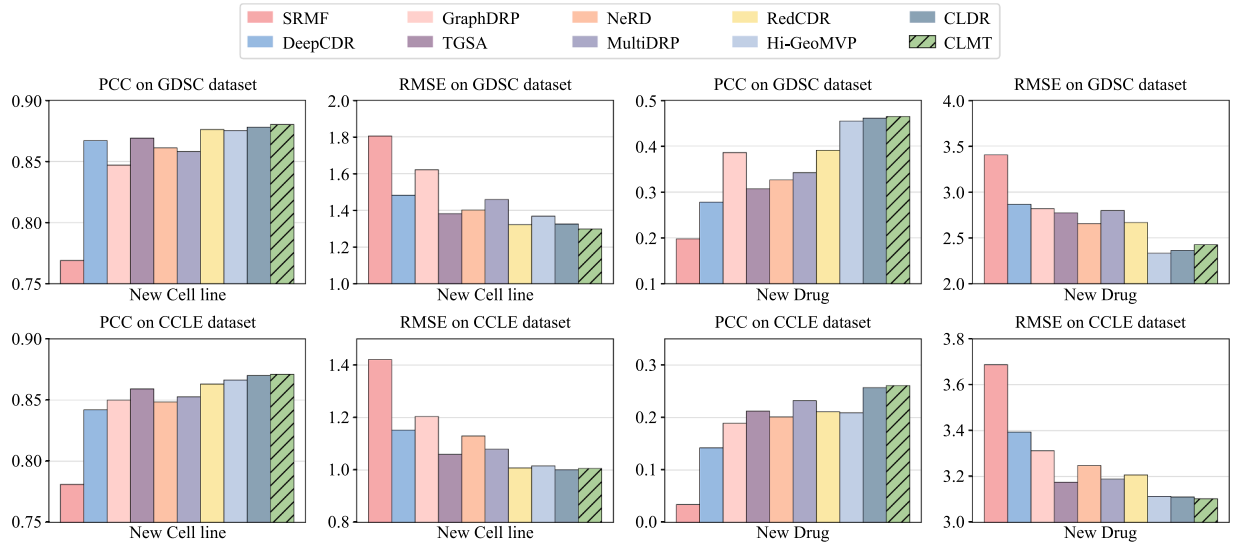


Fig. 4. Blind test performances for new cell lines and drugs..

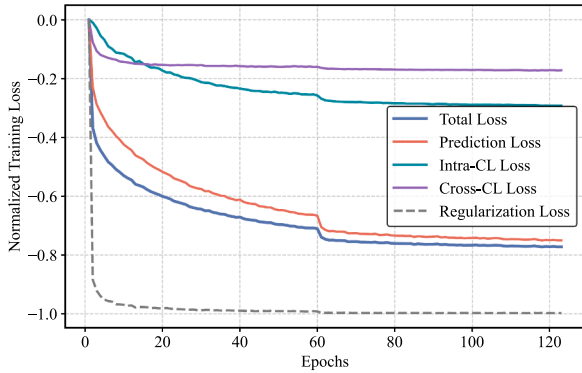


Fig. 5. Normalized training loss curves of different CLMT components..

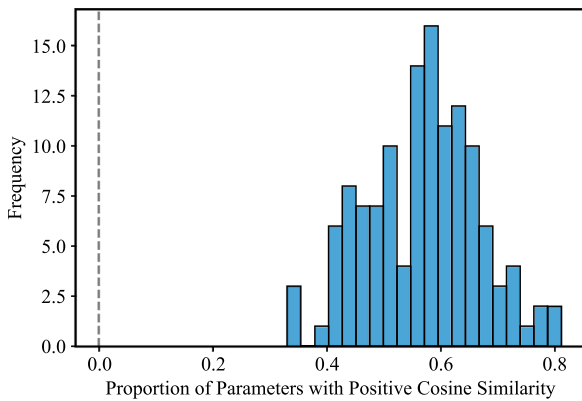


Fig. 6. Histogram of parameter-wise gradient alignment ratios.

their common latent subspace and align them within a unified representation space. In addition, prior theoretical work (Saunshi et al., 2019) establishes that contrastive objectives provide an upper bound on the expected risk of downstream predictors by learning semantically consistent representations. Our method employs complementary contrastive objectives where intra-modal learning enforces within-modality consistency and cross-modal learning aligns modalities into a unified semantic

space. Both objectives recover the same biological latent variable from different perspectives. These objectives serve as structural regularization constraining representation geometry, enabling joint optimization with task-specific prediction that identifies discriminative directions within this structured space.

In order to validate the effectiveness of multi-modal contrastive learning module, we present the normalized training loss of the joint optimization process in Fig. 5 on GDSC dataset. Each curve shows the normalized change relative to its initial value at epoch 0. This metric ranges from 0 to -1, where 0 represents the starting point and values approaching -1 indicate substantial loss reduction during training. From the Fig. 5, we observe a monotonic decrease tendency across all losses, indicating that each objective converges toward its optimum during joint optimization.

To assess the degree of gradient alignment between $\mathcal{L}_{DRP}$ and $\mathcal{L}_C$ throughout the training process, we calculate the proportion of parameters with positive cosine similarity between these two gradients at each epoch. Fig. 6 presents the histogram of these proportions on GDSC dataset. Following (Yu et al., 2020), a non-negative gradient cosine similarity indicates conflict-free multi-task optimization. The results show a predominance of values around 0.6, demonstrating that the majority of parameter consistently exhibit aligned gradient directions, confirming effective synergy between the contrastive learning auxiliary task and the main prediction task.

Notably, our method employs joint contrastive objectives that optimize for semantic correspondence rather than geometric point-wise alignment across modalities (Yuan et al., 2025). This optimization facilitates the discovery of a shared latent semantic subspace where representations from different modalities maintain structural coherence and can be seamlessly utilized by downstream task.

5.8. Clinical dataset validation

To further evaluate the translational potential and generalizability from *invitro* to *invivo* settings, we conduct external validation using The Cancer Genome Atlas (TCGA) bladder cancer dataset. Clinical drug response annotations are obtained following established protocols (Ding et al., 2016), where patient samples with “Complete Response” or “Partial Response” annotations are classified as drug-sensitive. Finally, we collect a cohort of 53 samples with complete three-omics profiles and

Table 5
Top-ranked Cell Lines for Selected Drugs.

Drug	Rank	Cell line	Disease	PMID
Temsilolimus	1	AGS	Gastric adenocarcinoma	N/A
	2	IHH-4	Thyroid gland papillary carcinoma	N/A
	3	VM-CUB-1	Bladder carcinoma	27235118 (Pinto-Leite et al., 2016)
	4	PC-3	Prostate carcinoma	18566209 (Garcia & Danielpour, 2008)
	5	LOXIMVI	Amelanotic melanoma	N/A
Vismodegib	1	OVCAR-3	High grade ovarian serous adenocarcinoma	31036756 (Kato, 2019)
	2	OAW-28	High grade ovarian serous adenocarcinoma	31036756 (Kato, 2019)
	3	NCI-H1355	Lung adenocarcinoma	31036756 (Kato, 2019)
	4	PANC-10-05	Pancreatic ductal adenocarcinoma	31036756 (Kato, 2019)
	5	TCCSUP	Bladder carcinoma	N/A

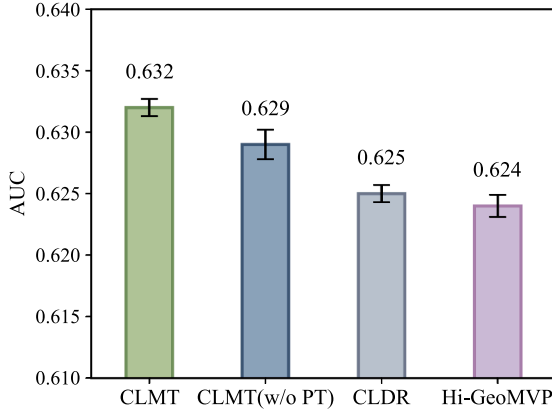


Fig. 7. Model performance on TCGA Bladder Cancer test set.

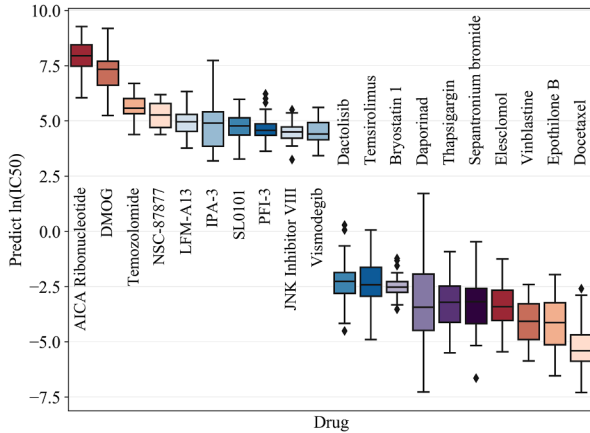


Fig. 8. The predicted $\ln(\text{IC}_{50})$ distribution of unknown responses that grouped by drugs.

drug response annotations, comprising 28 sensitive and 25 resistant cases.

We train the model on the GDSC dataset, with TCGA samples serving as an independent external validation cohort to assess cross-dataset transferability. Although the model is trained as a regression task, we evaluate performance using AUC. The predicted $\ln(\text{IC}_{50})$ values inherently rank samples by sensitivity, and AUC assesses whether this ranking correctly discriminates between sensitive and resistant samples. Our proposed model is compared against the competitive baselines, including CLDR, Hi-GeoMVP, and an ablation variant without pathway-level text, denoted as CLMT (w/o PT). The results are presented in Fig. 7. CLMT achieves superior performance with the highest AUC, while CLDR attains the inferior performance with an AUC of 0.625, representing a

modest improvement over Hi-GeoMVP. These cross-dataset validation results provide preliminary evidence of the effectiveness of incorporating biological text, particularly pathway-level descriptions, in enhancing multi-modal integration for DRP.

However, substantial performance gaps remain between *invitro* to *invivo* predictions. Recent advances in domain adaptation techniques, such as context-aware methods and knowledge transfer approaches, have shown promise for addressing *invitro* to *invivo* translation challenges (He et al., 2022; Liu & Li, 2025; Shubham et al., 2024). Future work will explore integrating textual information into these domain adaptation frameworks to further bridge the translational gap and enhance clinical applicability.

5.9. Case study

To evaluate CLMT's ability to identify clinically relevant cancer cell line-drug responses, we train the proposed model on known responses in the GDSC dataset and predict unknown responses. For visualization, we calculate the mean predicted $\ln(\text{IC}_{50})$ values grouped by drugs and show the distributions of the top-ranked and bottom-ranked 10 drugs using box plots in Fig. 8. As case studies, we select Temsirolimus and Vismodegib, with Table 5 presenting the top 5 predicted sensitive cell lines for Temsirolimus and resistant ones for Vismodegib.

Among the 20 drugs in Fig. 8, 5 are clinically approved: Docetaxel, Vinblastine, Temsirolimus, Vismodegib and Temozolomide. Docetaxel exhibits the lowest mean response value, indicating that it is generally more effective and likely demonstrates broad-spectrum activity. Indeed, it was approved by the FDA in 1996 and has established as an anticancer medication used to treat cancers of various origins (Knox et al., 2024; Montero et al., 2005). Vinblastine is also documented as a medication used to treat a wide variety of cancers (Knox et al., 2024).

As shown in Table 5, two cell lines VM-CUB-1 and PC-3 have been established as sensitive to Temsirolimus according to studies (Garcia & Danielpour, 2008; Pinto-Leite et al., 2016). For Vismodegib, previous review (Kato, 2019) indicates that genetic alterations or functional activation of the Hedgehog signaling pathway are relatively low frequency in breast, colorectal, gastric, pancreatic, non-small-cell lung, and ovarian cancers, therefore, Vismodegib as the Hedgehog Pathway Inhibitor, may exhibit limited efficacy in these cancer types. These findings demonstrate CLMT's ability to facilitate drug discovery in clinical trials.

6. Conclusion

In this work, we propose a novel multi-modal contrastive learning method based on dual-view molecular data and biological text. The primary contributions include the incorporation of molecular mechanism descriptions, as well as the design of intra- and cross-modal contrastive learning modules. Biomedical knowledge bridges the gap between molecular data and system-level biological understanding, leading to more informative and biologically meaningful representations. The intra-modal contrastive learning component preserves com-

plementary representations both attribute view and structure view molecular features, while cross-modal contrastive learning aligns molecular and textual representations, thereby facilitating the accurate prediction of drug response. Extensive experiments demonstrate that our method outperforms the competitive baselines on two commonly-used datasets. In future work, we will further explore the potential of textual knowledge to enhance model generalization in real-world clinical and novel drug discovery scenarios. In particular, LLMs represent a promising direction for in-depth biomedical knowledge integration and long-range semantic context understanding.

CRedit authorship contribution statement

Meiyu Duan: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Formal analysis, Conceptualization; **Xiaobo Li:** Software, Methodology; **Xiaodi Hou:** Visualization, Methodology; **Yanchen Qu:** Visualization, Validation; **Hai Cui:** Writing – original draft, Visualization, Software, Methodology; **Yijia Zhang:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization.

Data availability

Source code and datasets are available at this [github link](#).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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